Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/23/2006
PI Name:	Cohen, Richard J. M.D., Ph.D.		
Project Title:	Cardiovascular Effects of Simulated Microgravity in Man		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02142-1323	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	10/01/2000	**End Date:**	02/29/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	CA.002.02		
Performance Goal No.:			
Performance Goal Text:			

Many astronauts after being weightless in space become hypotensive and presyncopal upon assuming an upright position. This phenomenon, known as orthostatic intolerance, may interfere with astronaut function during reentry and following spaceflight, and may limit the ability of an astronaut to exit a landed spacecraft unaided during an emergency. Orthostatic intolerance is more pronounced following long-term spaceflight (1) and is a major concern with respect to the extended flights expected aboard the International Space Station and for interplanetary exploration class missions, such as a human mission to Mars. This problem has also been observed to be more pronounced among women than among men (2). In addition to the problem of post-flight orthostatic intolerance, a variety of heart rhythm disturbances have been observed in astronauts during and after space flight (3-6). The potential lethal arrhythmic risk for astronauts is sustained ventricular tachycardia or ventricular fibrillation, while non-sustained ventricular tachycardia could cause syncope. Older individuals, particularly men, may be most susceptible to ventricular arrhythmias during flight.
In previous ground based bed-rest studies sponsored by NSBRI we have applied two new techniques that we have

Task Description:	developed to study the effects of simulated microgravity on the cardiovascular system. Cardiovascular system identification (CSI) has been used a non-invasive means of measuring alterations in closed-loop cardiovascular regulation and the measurement of microvolt level T wave alternans (TWA)has been used as a non-invasive measure of susceptibility to ventricular arrhythmias. We have also successfully tested the alpha-1 sympathetic agonist midodrine as a countermeasure to the development of orthostatic intolerance. We have found that 16 days of bed-rest results in altered cardiovascular regulation in particular alterations in baroreceptor sensitivity, altered electrical stability of the heart, and that midodrine is an effective countermeasure to the development of orthostatic hypotension. In this project we apply the same measurement techniques of CSI and TWA to two groups of subjects before and after 16 days of bedrest. Premenopausal women and men over age 55. The women will be randomized to placebo or midodrine to see if this countermeasure is effective in these subjects who are more susceptible to orthostatic hypotension than men of the same age. The group of older men will be randomized to placebo or spironolactone, an aldosterone blocking agent. Spironolactone will be evaluated as a countermeasure to the pro-arrhythmic action of aldosterone which is elevated in these subjects during bed rest.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	To conduct these studies we have developed two new technologies: the measurement of Microvolt T Wave Alternans (MTWA) and Cardiovascular System Identification (CSI). MTWA has already been commercialized, cleared by the FDA a non-invasive predictor of cardiac arrest and sudden cardiac death, and has been issued a reimbursement code by Medicare. It is in widespread and growing clinical use, and is already having an impact in reducing mortality from sudden cardiac death - the cause of one in seven of all deaths in the United States. Cardiovascular System Identification is a powerful non-invasive technique for assessing closed loop cardiovascular regulation and autonomic activity. It has applications as a diagnostic technique, and as a means to guide therapy, for a wide range of disorders including heart failure, hypertension and diabetes.
Task Progress:	No progress report this reporting period.
Bibliography Type:	Description: (Last Updated: 01/18/2008)